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Inventor(s)

YAMADA; Kazuyuki

DISPLAY DEVICE AND REPAIR METHOD FOR DISPLAY DEVICE

Abstract

A display device includes a first LED chip arranged in a pixel, a first mounting pad arranged in the pixel and connected to the first LED chip, a second mounting pad arranged in the pixel, overlapping the first mounting pad in a plan view and electrically connected to the second mounting pad, and an insulating layer between the first mounting pad and the second mounting pad.

Inventors: YAMADA; Kazuyuki (Tokyo, JP)

Applicant: Japan Display Inc. (Tokyo, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2024-033572, filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] One embodiment of the present invention relates to a display device and a repair method for a display device. In particular, the present invention relates to a display device with an LED (Light Emitting Diode) chip and a repair method for the display device.

BACKGROUND

[0003] Recently, an LED display in which a minute LED chip is mounted on each pixel is being developed as a next-generation display device. Normally, an LED display has a structure in which a plurality of LED chips is mounted on a circuit substrate constituting a pixel array. The circuit substrate has a drive circuit for causing an LED to emit light at a position corresponding to each pixel. These drive circuits are electrically connected to each LED chip, respectively.

[0004] There are various methods for mounting a plurality of LED chips on a circuit substrate. For example, there is the well-known method of removing only the support substrate after bonding the LED chips on the support substrate to the circuit substrate. For example, Japanese laid-open patent publication No. 2023-005830 describes a technique for selectively irradiating laser light to LED chips at a high throughput by using a light-shielding mask.

[0005] In a display device with mounted LED chips, there is a need to improve the yield rate when mounting the LED chips from the support substrate to the circuit substrate. For example, a connection failure between the LED chip and the circuit substrate will reduce the yield. Such a connection failure is undesirable since it causes defects in the pixel.

SUMMARY

[0006] A display device in an embodiment according to the present invention includes a first LED chip arranged in a pixel, a first mounting pad arranged in the pixel and connected to the first LED chip, a second mounting pad overlapping and electrically connected to the first mounting pad in a plan view, and an insulating layer between the first mounting pad and the second mounting pad.

[0007] A repair method for a display device in an embodiment according to the present invention, the method includes irradiating a first laser light to a first LED chip determined to be defective in a pixel and removing the first LED chip, irradiating a second laser light to a first mounting pad connected to the first LED chip in the pixel and removing the first mounting pad, irradiating a third laser light to an insulating layer overlapping the first mounting pad and removing the insulating layer, and exposing a second mounting pad overlapping the first mounting pad in a plan view, and mounting a second LED chip on the second mounting pad.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a plan view showing a schematic configuration of a display device according to an embodiment of the present invention.

[0009] FIG. 2 is a block diagram showing a circuit configuration of the display device according to an embodiment of the present invention.

[0010] FIG. 3 is an example of a circuit diagram showing a configuration of a pixel circuit of the display device according to an embodiment of the present invention.

[0011] FIG. 4 is a plan view showing a schematic configuration of a pixel in a display device according to an embodiment of the present invention.

[0012] FIG. 5 is a cross-sectional view showing a schematic configuration of a pixel in a display device according to an embodiment of the present invention.

[0013] FIG. 6 is a plan view showing a schematic configuration of a pixel in a display device

according to an embodiment of the present invention.

[0014] FIG. **7** is a cross-sectional view showing a schematic configuration of a pixel in a display device according to an embodiment of the present invention.

[0015] FIG. **8** is a flowchart showing a repair method for a display device according to an embodiment of the present invention.

[0016] FIG. **9** is a cross-sectional view showing a repair method for a display device according to an embodiment of the present invention.

[0017] FIG. **10** is a cross-sectional view showing a repair method for a display device according to an embodiment of the present invention.

[0018] FIG. **11** is a cross-sectional view showing a repair method for a display device according to an embodiment of the present invention.

[0019] FIG. **12** is a cross-sectional view showing a repair method for a display device according to an embodiment of the present invention.

[0020] FIG. **13** is a cross-sectional view showing a repair method for a display device according to an embodiment of the present invention.

[0021] FIG. **14** is a plan view showing a schematic configuration of a repaired pixel in a display device according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0022] Embodiments of the present invention will be described below with reference to the drawings and the like. However, the present invention can be implemented in various modes without departing from the gist thereof. The present invention is not to be construed as being limited to the description of the following exemplary embodiments. For the sake of clarity of description, the drawings may be schematically represented with respect to widths, thicknesses, shapes, and the like of the respective portions in comparison with actual modes. However, the drawings are merely examples and do not limit the interpretation of the present invention.

[0023] In describing the present embodiment, a direction from a substrate to an LED chip is “above” and the opposite direction is “below”. However, the expression “above” or “below” merely describes the upper limit relationship of each element. For example, the expression that an LED chip is arranged above a substrate also includes the presence of other members between the substrate and the LED chip. Furthermore, the expression “above” or “below” includes not only the case where chips overlap but also the case where they do not overlap in a plan view.

[0024] In the description of the embodiment of the present invention, elements having the same functions as those of the elements already described are denoted by the same symbols or the same symbols with symbols such as letters of the alphabet, and the description thereof may be omitted. For example, in the case where an element needs to be described separately for each color of RGB, it may be distinguished by the symbols R, G, or B after the symbols indicating the element. On the other hand, in the case where the elements need not be described separately for each color of RGB, only a symbol indicating the elements will be used.

First Embodiment

[0025] A display device **10** according to one embodiment of the present invention will be described [Configuration of Display Device]

[0026] FIG. **1** is a plan view showing a schematic configuration of a display device according to an embodiment of the present invention. The display device **10** has a circuit substrate **100**, a flexible printed circuit substrate **160** (FPC **160**), and an IC chip **170** as shown in FIG. **1**. The display device **10** has a display area **112**, a peripheral area **114**, and a terminal area **116**.

[0027] The display area **112** is an area in which a plurality of pixels **110** including an LED chip **202** is arranged in a row direction (D1 direction) and a column direction (D2 direction). Specifically, in the present embodiment, a pixel **110R** including a red LED element **202R**, a pixel **110G** including a green LED element **202G**, and a pixel **110B** including a blue LED element **202B** are arranged in the area. FIG. **1** shows an example in which the pixel **110R**, pixel **110G**, and pixel **110B** are sub-

pixels, and they constitute a single pixel **110**. The display area **112** functions as an area for displaying an image corresponding to a video signal.

[0028] The peripheral area **114** is an area around the display area **112**. The peripheral area **114** is an area provided with driver circuits (a data driver circuit **130** and a gate driver circuit **140** shown in FIG. 2) for controlling pixel circuits (pixel circuits **120** shown in FIG. 2) provided in each pixel **110**.

[0029] The terminal area **116** is an area in which a plurality of wirings connected to the aforementioned driver circuits is aggregated. The flexible printed circuit substrate **160** is electrically connected to a plurality of wirings in the terminal area **116**. A video signal (data signal) or control signal output from an external device (not shown) is input to the IC chip **170** via the wiring (not shown) provided in the flexible printed circuit substrate **160**. The IC chip **170** performs various signal processing on the video signal and generates a control signal required for display control. The video signal and control signal output from the IC chip **170** are input to the display device **10** via the flexible printed circuit substrate **160**.

[Circuit Configuration of Display Device **10**]

[0030] FIG. 2 is a block diagram showing a circuit configuration of the display device **10** according to one embodiment of the present invention. As shown in FIG. 2, the display area **112** is provided with the pixel circuit **120** corresponding to each pixel **110**. In the present embodiment, the pixel circuit **120R**, the pixel circuit **120G**, and the pixel circuit **120B** are provided corresponding to the pixel **110R**, the pixel **110G**, and the pixel **110B**, respectively. That is, a plurality of pixel circuits **120** is arranged in the row direction (D1 direction) and the column direction (D2 direction) in the display area **112**.

[0031] FIG. 3 is a circuit diagram showing a configuration of the pixel circuit **120** of the display device **10** according to one embodiment of the present invention. The pixel circuit **120** is arranged in an area surrounded by a data line **121**, a gate line **122**, an anode power line **123**, and a cathode power line **124**. The pixel circuit **120** of the present embodiment includes a select transistor **126**, a drive transistor **127**, a storage capacity **128**, and an LED **129**. The LED **129** corresponds to the LED chip **202** shown in FIG. 1. In the pixel circuit **120**, circuit elements other than the LED **129** correspond to a drive circuit **102** provided in the circuit substrate **100**. That is, the pixel circuit **120** is completed with the LED chip **202** mounted on the circuit substrate **100**.

[0032] As shown in FIG. 3, a source electrode, gate electrode, and drain electrode of the select transistor **126** are connected to the data line **121**, the gate line **122**, and a gate electrode of the drive transistor **127**, respectively. The source electrode, gate electrode, and drain electrode of the drive transistor **127** are connected to the anode power line **123**, a drain electrode of the select transistor **126**, and the LED **129**, respectively. The storage capacity **128** is connected between the gate electrode and the drain electrode of the drive transistor **127**. That is, the storage capacity **128** is connected to the drain electrode of the select transistor **126**. The anode and cathode of the LED **129** are connected to the drain electrode of the drive transistor **127** and the cathode power line **124**, respectively.

[0033] A gradation signal for determining an emission intensity of the LED **129** is supplied to the data line **121**. A gate signal for selecting the select transistor **126** that writes the gradation signal is supplied to the gate line **122**. When the select transistor **126** is turned on, the gradation signal is stored in the storage capacity **128**. Thereafter, when the drive transistor **127** is turned on, a drive current corresponding to the gradation signal flows through the drive transistor **127**. When the drive current output from the drive transistor **127** is input to the LED **129**, the LED **129** emits light with a light emission intensity corresponding to the gradation signal.

[0034] Referring to FIG. 2 again, a data driver circuit **130** is arranged at a position adjacent to the display area **112** in the column direction (D2 direction). A gate driver circuit **140** is arranged at a position adjacent to the display area **112** in the row direction (D1 direction). In the present embodiment, although two gate driver circuits **140** are provided on both sides of the display area

112, only one of them may be provided.

[0035] Both the data driver circuit **130** and the gate driver circuit **140** are arranged in the peripheral area **114**. However, the area where the data driver circuit **130** is arranged is not limited to the peripheral area **114**. For example, the data driver circuit **130** may be arranged on the flexible printed circuit substrate **160**.

[0036] The data line **121** shown in FIG. 3 extends from the data driver circuit **130** in the D2 direction and is connected to the source electrode of the select transistor **126** in each pixel circuit **120**. The gate line **122** extends from the gate driver circuit **140** in the D1 direction and is connected to the gate electrode of the select transistor **126** in each pixel circuit **120**.

[0037] A terminal portion **150** is arranged in the terminal area **116**. The terminal portion **150** is connected to the data driver circuit **130** via a connecting wiring **151**. Similarly, the terminal portion **150** is connected to the gate driver circuit **140** via a connecting wiring **152**. In addition, the terminal portion **150** is connected to the flexible printed circuit substrate **160**.

[Configuration of Pixel **110**]

[0038] FIG. 4 is a plan view showing a pixel of a display device according to an embodiment of the present invention. An LED chip **202**, a first mounting pad **25a-1**, a second mounting pad **25a-2**, and a conductive layer **30** are arranged in the pixel **110**. As shown in FIG. 4, the first mounting pad **25a-1** and the second mounting pad **25a-2** are arranged corresponding to each of the red LED chip **202R**, the green LED chip **202G**, and the blue LED chip **202B**.

[0039] The mounting pad in the present embodiment is an area formed by a metal film for mounting the LED chip on the substrate, and is an electrode. The shape of the mounting pad in a plan view is optional, and may be designed as needed to adapt to the electrode shape and electrode arrangement of the LED chip.

[0040] The LED chip **202** overlaps the first mounting pad **25a-1**, the second mounting pad **25a-2**, and the conductive layer **30** in a plan view. In a plan view, the area occupied by the LED chip **202** is smaller than the area of the first mounting pad **25a-1**, the second mounting pad **25a-2**, and the conductive layer **30**. The larger area of the first mounting pad **25a-1**, the second mounting pad **25a-2**, and the conductive layer **30** relative to the area of the LED chip **202** can provide a margin of alignment accuracy when mounting the LED chip **202** on the pad.

[0041] The first mounting pad **25a-1** may have a region that overlaps the LED chip **202** and the second mounting pad **25a-2** and a region that does not overlap the LED chip **202**.

[0042] A new LED chip **402** (not shown) may be mounted in the region where the LED chip **202** is mounted in place of the currently mounted LED chip **202**, as described below in detail. When mounting the new LED chip **402**, the LED chip **202** is removed and the first mounting pad **25a-1** is removed to further expose the top surface of the second mounting pad **25a-2**.

[0043] A connection portion is provided to electrically connect the first mounting pad **25a-1** and the second mounting pad **25a-2** in the region where the LED chip **202** does not overlap the first mounting pad **25a-1** and the second mounting pad **25a-2**. As shown in FIG. 4, the first mounting pad **25a-1** is connected to the second mounting pad **25a-2** through a contact hole **262** arranged in the region where the first mounting pad **25a-1** and the second mounting pad **25a-2** overlap. By connecting the first mounting pad **25a-1** and the second mounting pad **25a-2**, the same potential can be provided to the first mounting pad **25a-1** and the second mounting pad **25a-2**.

[0044] The first mounting pad **25a-1** may be formed to surround the LED chip **202**. The first mounting pad **25a-1** has a portion surrounding the LED chip **202** and a portion extending from the portion surrounding the LED chip **202** toward an outer edge **110e** of the pixel **110**, as shown in FIG. 4. Here, the portion surrounding the LED chip **202** corresponds to the region overlapping the LED chip **202** and the second mounting pad **25a-2** of the first mounting pad **25a-1** as described above, and the portion extending from the portion surrounding the LED chip **202** to the outer edge **110e** of the pixel **110** corresponds to the region of the first mounting pad **25a-1** that does not overlap the LED chip **202**.

[0045] The area of the first mounting pad **25a-1** may be smaller than the area of the second mounting pad **25a-2**. When a plurality of red LED chips **202R**, green LED chips **202G**, and blue LED chips **202B** are provided in the pixel **110**, as shown in FIG. **4**, for example, since area of the first mounting pad **25a-1** is small, the distance between the first mounting pads **25a-1** connecting to each of the plurality of LED chips **202** becomes longer. The longer distance makes it easier to remove the first mounting pad **25a-1** connecting to one of the plurality of LED chips **202**.

However, depending on the arrangement of the plurality of LED chips **202** in the pixel **110**, the area of the first mounting pad **25a-1** may be larger than the area of the second mounting pad **25a-2**.

[0046] The second mounting pad **25a-2** may be formed to surround the LED chip **202** and the first mounting pad **25a-1**. The second mounting pad **25a-2** may have a portion surrounding the LED chip **202** and a portion extending from the portion surrounding the LED chip **202** to the outer edge **110e** of the pixel **110**, as shown in FIG. **4**. The portion surrounding the LED chip **202** corresponds to a repair region RP, the portion surrounding the LED chip **202** and the portion extending from the portion surrounding the LED chip **202** to the outer edge **110e** of the pixel **110** corresponds to a region OT.

[0047] The repair region RP is a region where a new LED chip **302** is mounted in place of the currently mounted LED chip **202** when the LED chip **202** is determined to be defective. The region OT is a region where a portion to be connected to the first mounting pad **25a-1** is arranged.

[0048] The first mounting pad **25a-1** and the second mounting pad **25a-2** are arranged to be separated from adjacent first mounting pads **25a-1** and second mounting pads **25a-2**. When a plurality of LED chips **202** are arranged in the pixel **110**, the first mounting pad **25a-1** and the second mounting pad **25a-2**, which connect to each of the plurality of

[0049] LED chips **202**, are arranged at a distance. As shown in FIG. **4**, the first mounting pad **25a-1** and the second mounting pad **25a-2**, which overlap and are electrically connected with the red LED chip **202R**, are arranged to be separated from the first mounting pad **25a-1** and the second mounting pad **25a-2**, which overlap and are electrically connected with the adjacent green LED chip **202G**.

[0050] The conductive layer **30** may be electrically connected to the LED chip **202**. The conductive layer **30** is arranged to overlap the LED chip **202**, the first mounting pad **25a-1**, and the second mounting pad **25a-2**. The conductive layer **30** may be formed to surround the LED chip **202**. As shown in FIG. **4**, the conductive layer **30** may have a portion surrounding the LED chip **202** and a portion extending from the portion surrounding the LED chip **202** toward the outer edge **110e** of the pixel **110**. The portion of the conductive layer **30** extending toward the outer edge **110e** of the pixel **110** connects with the conductive layer **30** which electrically connects to the other LED chips **202** when multiple LED chips **202** are provided in the pixel **110**. For example, as shown in FIG. **4**, the conductive layer **30** that overlaps and is electrically connected with the red LED chip **202R** is connected to the conductive layer **30** that overlaps and is electrically connected with the green LED chip **202G** at a portion extending toward the outer edge **110e** of the pixel **110**. As such, it is possible to provide the same potential to those conductive layers **30** by connecting the conductive layers **30** electrically connected to the plurality of LED chips **202** in the same pixel **110** to each other.

[0051] FIG. **5** is a cross-sectional view showing a configuration of the pixel in a display device according to one embodiment of the present invention. Specifically, FIG. **5** is a cross-sectional view along the A1-A2 line in FIG. **4**. In FIG. **5**, the same reference number is attached to the same configuration of the pixel **110** shown in FIG. **4**, and redundant explanations are omitted.

[0052] The pixel **110** has the drive transistor **127** provided on an insulating substrate **11**. A glass substrate or a transparent substrate with an insulating layer on a resin substrate may be used as the insulating layer **11**.

[0053] The drive transistor **127** has a semiconductor layer **12**, a gate insulating layer **13**, and a gate electrode **14**. A source electrode **16** and a drain electrode **17** are connected to the semiconductor layer **12** via an insulating layer **15**. Although not shown, the gate electrode **14** is connected to a

drain electrode of the select transistor **126** shown in FIG. 3.

[0054] A wiring **18** is provided in the same layer as the source electrode **16** and the drain electrode **17**. The wiring **18** functions as the anode power line **123** shown in FIG. 3. Therefore, the source electrode **16** and the wiring **18** are electrically connected by a connecting wiring **20** provided on a planarization layer **19**. The planarization layer **19** is a transparent resin layer using a resinous material such as polyimide or acrylic. The connecting wiring **20** is a transparent conductive layer using a metal oxide material such as ITO. However, the present invention is not limited to this example, and other metal materials may be used as the connecting wiring **20**.

[0055] An insulating layer **21** composed of silicon nitride or the like is provided on the connecting wiring **20**. A conductive layer **22** and a conductive layer **23** are provided on the insulating layer **21**. In the present embodiment, the conductive layer **22** and the conductive layer **23** are electrodes consisting of a light-shielding metallic material. The conductive layer **22** is connected to the drain electrode **17** through contact hole(s) (opening(s)) in the planarization layer **19** and the insulating layer **21**.

[0056] A planarization layer **24** is provided on the conductive layer **22** and the conductive layer **23**. The planarization layer **24** is a transparent resin layer composed of polyimide, acrylic, or other resin material.

[0057] The second mounting pad **25a-2** is formed on the planarization layer **24**. The second mounting pad **25a-2** is connected to the conductive layer **22** through a contact hole **264** in the planarization layer **24**.

[0058] An insulating layer **28** is provided on the second mounting pad **25a-2**. The thickness of the insulating layer **28** is greater than or equal to 1 μm and less than or equal to 2 μm . The insulating layer **28** is an organic insulating layer composed of a photosensitive resin material or the like. The organic insulating layer having a thickness greater than 1 μm and less than 2 μm used for the insulating layer **28** provides ease of removal by laser or other means.

[0059] The first mounting pad **25a-1** and a mounting pad **25b** are provided on the insulating layer **28**. The first mounting pad **25a-1** is arranged to overlap the second mounting pad **25a-2** in a cross-sectional view. The insulating layer **28** is arranged between the first mounting pad **25a-1** and the second mounting pad **25a-2**.

[0060] The first mounting pad **25a-1** is connected to the second mounting pad **25a-2** through the contact hole **262** provided in the insulating layer **28**. The first mounting pad **25a-1** connecting with the second mounting pad **25a-2** can provide the same potential as described above. The second mounting pad **25a-2** is at the same potential as the first mounting pad **25a-1** and is arranged in the same pixel **110** so that the second mounting pad **25a-2** can function as a spare mounting pad for the first mounting pad **25a-1**. Furthermore, the second mounting pad **25a-2** is arranged to overlap the first mounting pad **25a-1**, and the first mounting pad **25a-1** is removed so that the second mounting pad **25a-2** can be used as a spare mounting pad for the first mounting pad **25a-1**.

[0061] The mounting pad **25b** is connected to the conductive layer **23** through the contact holes **263** in the planarization layer **24** and the insulating layer **28**. In FIG. 5, although the connection between the mounting pad **25b** and the conductive layer **23** is through the contact hole **263** only, they may be electrically connected through a plurality of contact holes, as in the connection between the first mounting pad **25a-1** and the conductive layer **22**. The first mounting pad **25a-1**, the second mounting pad **25a-2**, and the mounting pad **25b** are composed of metallic materials such as aluminum, titanium, molybdenum, tantalum, tungsten, etc.

[0062] The LED chip **202** is provided on the first mounting pad **25a-1**. The LED chip **202** is arranged to overlap the first mounting pad **25a-1**, the second mounting pad **25a-2**, and the insulation layer **28**. The LED chip **202** is arranged so as not to overlap the contact hole **262** and the contact hole **264**. The contact hole **262** may be arranged between the LED chip **202** and the contact hole **264**. The contact hole **262** is arranged between the LED chip **202** and the contact hole **264** without the LED chip **202** overlapping the contact hole **264** and the contact hole **262** so that when

installing a new LED chip **402** (not shown) in place of the LED chip **202**, damage to the contact holes **264** due to removal of the LED chip **202** can be prevented.

[0063] The LED chip **202** corresponds to the LED **129** in the circuit diagram shown in FIG. **3**. An anode **303a** of the LED chip **202** is connected to the conductive layer **22** connected to the drain electrode **17** of the drive transistor **127**. A cathode **203b** of the LED chip **202** is connected to the conductive layer **23**. The conductive layer **23** is electrically connected to the cathode power line **124** shown in FIG. **3**. The arrangement of the anode **203a** and cathode **203b** shown in FIG. **5** is an example, and the arrangement of the anode **203a** and cathode **203b** may be reversed.

[0064] A first connecting electrode **103a** is provided between the first mounting pad **25a-1** and the LED chip **202**. In the present embodiment, an electrode composed of tin (Sn) is arranged as the first connecting electrode **103a**.

[0065] The first connecting electrode **103a** is bonded to the anode **203a** of the LED chip **202**. The anode **203a** is provided on one of the surfaces of the LED chip **202**. On the other hand, the cathode **203b** is electrically connected to the conductive layer **30** on the other surface of the LED chip **202**. The conductive layer **30** functions as the connecting electrode of the LED chip **202** and is electrically connected to the conductive layer **23**. In the present embodiment, the anode **203a** and the cathode **203b** may be electrodes composed of gold (Au). The first connecting electrode **103a** and the anode **203a** of the LED chip **202** may be joined to each other by a heat treatment with laser light irradiation. As the laser light, a laser light that is absorbed by the first connecting electrode **103a** or the anode **203a** may be selected.

[0066] A planarization layer **26** is provided over the first mounting pad **25a-1** and the mounting pad **25b**. The planarization layer **26** covers the entire side surface of the LED chip **202**. The conductive layer **30** is connected to the mounting pad **25b**, which is connected to the conductive layer **23** through the contact hole **261** in the planarization layer **26**. The planarization layer **26** is a transparent resin layer composed of an insulating resin material such as polyimide, acrylic, or the like.

[0067] As described above, the display device **10** according to the present embodiment includes the LED chip **202** arranged in the pixel **110**, the first mounting pad **25a-1** arranged in the pixel **110** and connected to the LED chip **202**, the second mounting pad **25a-2** arranged in the pixel **110** and overlapping the first mounting pad **25a-1** in a plan view and electrically connected to the second mounting pad **25a-2**, and the insulating layer between the first mounting pad **25a-1** and the second mounting pad **25a-2**. In the pixel **110** where the first mounting pad **25a-1** is arranged, the second mounting pad **25a-2** is overlapped by and electrically connected to the first mounting pad **25a-1**, so that the second mounting pad **25a-2** in the pixel **110** can be provided as a spare mounting pad without providing an additional region for providing the second mounting pad **25a-2**. In the display device **10**, the small proprietary area per pixel allows more pixels to be provided in the display area **112**, thereby providing a high-definition display device.

<Variation>

[0068] In one embodiment of the present invention described above, the LED chip has a structure in which a semiconductor layer and an active layer, not shown, are sandwiched by a pair of the anode **203a** and the cathode **203b** in the circuit substrate **100**, the so-called vertical structure. However, the connection between the LED chip and the connecting electrode on the side of the insulating substrate **11** is not limited to the connection with a vertical structure.

[0069] FIG. **6** is a plan view showing the configuration of a pixel **110a** of the display device **10** according to a variation of one embodiment of the present invention. FIG. **7** is a cross-sectional view showing the configuration of the pixel **110a** of the display device **10** according to a variation of one embodiment of the present invention. As shown in FIG. **7**, in the pixel **110a** related to the variant example, the LED chip **202aR** has an anode **203a** electrically connected to the conductive layer **22** and a cathode **203b** electrically connected to the conductive layer **23** on one side, and each is electrically connected to the first mounting pad **25a-1** and second mounting pad **25b-1**, in a so-

called flip chip mounting method. In FIGS. 6 and 7, the same reference numbers are attached to the same configuration of the pixel **110** shown in FIGS. 4 and 5, and redundant explanations are omitted.

[0070] The LED chip **202a**, the first mounting pad **25a-1**, and the second mounting pad **25a-2** are arranged in the pixel **110a**. A plurality of LED chips **202** can be provided in the pixel **110a**, and a plurality of LED chips **202aR**, LED chips **202aG**, and LED chips **202aB** is arranged in the pixel **101a** as shown in FIG. 6.

[0071] The LED chip **202a** overlaps the first mounting pad **25a-1**, the second mounting pad **25a-2**, the first mounting pad **25b-1** and the second mounting pad **25b-2** in a plan view.

[0072] The first mounting pad **25a-1** overlaps the second mounting pad **25a-2** in a plan view. The first mounting pad **25b-1** overlaps the second mounting pad **25b-2** in a plan view. The first mounting pad **25b-1** may be connected to the second mounting pad **25b-2** in a region not overlapping the LED chip **202a**. The first mounting pad **25b-1** may have a portion extending from the portion that overlaps the LED chip **202a** to the outer edge **110e** of the pixel **110**. The portion extending to the outer edge **110e** of the pixel **110** of the first mounting pad **25b-1** corresponds to a region which does not overlap the LED chip **202a** as shown in FIG.

[0073] **6**. The first mounting pad **25b-1** may be connected through the contact hole **267**, which is arranged in a region not overlapping the LED chip **202a**. The second mounting pad **25b-2** can be connected to the first mounting pad **25b-1** to provide the same potential as the first mounting pad **25b-1**. The second mounting pad **25b-2** is at the same potential as the first mounting pad **25b-1** and is arranged in the same pixel **110** so that the second mounting pad **25a-2** **1** can function as a spare mounting pad for the first mounting pad **25b-1**. Furthermore, the second mounting pad **25b-2** is arranged to overlap the first mounting pad **25b-1**, so that it allows an LED chip to be mounted on the same position where the first mounting pad **25b-1** was mounted by removing the first mounting pad **25b-1**.

[0074] The pixel **110a** has the drive transistor **127** provided on the insulating substrate **11**. The structure of the drive transistor **127** shown in FIG. 7 is the same as the drive transistor **127** shown in FIG. 4. The wiring **18** is provided in the same layer as the source electrode **16** and drain electrode **17** of the drive transistor **127**. The planarization layer **19** is provided on the source electrode **16**, the drain electrode **17** and the wiring **18**. The source electrode **16** and the wiring **18** are electrically connected by the connecting wiring **20** provided on the planarization layer **19**.

[0075] The insulating layer **21** composed of silicon nitride or the like is provided above the connecting wiring **20**, and the conductive layer **22** and the conductive layer **23** are provided on the insulating layer **21**. The conductive layer **22** is connected to the drain electrode **17** through a contact hole(s) (opening(s)) in the planarization layer **19** and the insulating layer **21**.

[0076] The conductive layer **22** and the conductive layer **23** are connected to the second mounting pad **25a-2** and the second mounting pad **25b-2** through the contact hole **264** and the contact hole **265** in the planarization layer **24**, respectively. The second mounting pad **25b-2** may be configured in the same manner as the second mounting pad **25a-2**.

[0077] The insulating layer **28** is provided on the second mounting pad **25a-2** and the second mounting pad **25b-2**. The first mounting pad **25a-1** and the first mounting pad **25b-1** are provided on the insulating layer **28**. The first mounting pad **25a-1** and the first mounting pad **25b-1** are connected to the second mounting pad **25a-2** and the second mounting pad **25b-2** through the contact hole **262** and the contact hole **267** provided in the insulating layer **28**, respectively. The first mounting pad **25b-1** may be configured in the same manner as the first mounting pad **25a-1**.

[0078] The first mounting pad **25a-1**, which is provided on the insulating layer **28** and electrically connects to the anode **203a**, and the first mounting pad **25b-1**, which is provided on the insulating layer **28** and is electrically connected to the cathode **203b**, correspond to a first p-electrode side pad **25 a-1** and a first n-electrode side pad **25b-1** corresponding to the LED chip **202a**. The second mounting pad **25a-2**, which is provided on the planarization layer **24** and is electrically connected

to the anode **203a**, and the second mounting pad **25b-2**, which is provided on the planarization layer **24** and is electrically connected to the cathode **203b**, correspond to a second p-electrode side pad **25a-2** and a second n-electrode side pad **25b-2** corresponding to the LED chip **202a**.
[0079] The first connecting electrode **103a** is provided between the first p-electrode side pad **25a-1** and the anode **203a**. The first connecting electrode **103b** is provided between the first n-electrode side **25b-1** and cathode **203b**. The first connecting electrode **103b** may be configured in the same manner as the first mounting pad **25a-1**.

Second Embodiment

[0080] The repair method for the display device **10** equipped with the pixel **110** described in the first embodiment will be described. The description may omit descriptions of configurations identical or similar to those described in the first embodiment.

[Repair Method for Display Device]

[0081] FIG. **8** is a flowchart diagram showing a repair method for a display device according to an embodiment of the present invention. Specifically, FIG. **8** shows the process of removing the LED chip determined to be defective from among a plurality of LED chips **202** formed on the circuit substrate **100** and mounting a new LED chip. FIGS. **9** to **13** are cross-sectional views showing a repair method for a display device according to an embodiment of the present invention. FIGS. **9** to **13** are cross-sectional views showing a repair method for a display device. In FIGS. **9** to **13**, the configuration of the pixel **110** shown in FIG. **5** is shown in a simplified form. The process of removing an LED chip determined to be defective from the circuit substrate **100** and mounting a new LED chip on the circuit substrate **100** is described here while referring to FIG. **8**.

[0082] First, in step **S11** of FIG. **8**, the red LED chip **202R** that is arranged in the pixel **110** and determined to be defective is removed. For example, if the red LED chip **202R** shown in FIG. **4** is determined to be defective, the red LED chip **202R** arranged in the pixel **110** and determined to be defective is irradiated with a laser light **50E**, and the red LED chip **202R** is removed from the pixel **110**.

[0083] The laser light **50E** can be selected as a laser light that can be absorbed by the red LED chip **202R**. In the present embodiment, for example, ultraviolet light may be used as the laser light **50E**. An excimer laser or the like may be used as the light source of the laser light **50E**.

[0084] Next, in step **S12** of FIG. **8**, the first mounting pad **25a-1** to which the red LED chip **202R** was connected is removed. For example, the first mounting pad **25a-1** to which the red LED chip **202R** was connected is irradiated with a laser light **50P** to remove the first mounting pad **25a-1** from the pixel **110**. This process exposes the insulating layer **28** that was overlapped by the red LED chip **202R** that was determined to be defective. As shown in FIG. **10**, the first mounting pad **25a-1** to which the red LED chip **202R** was connected is irradiated with the laser light **50P** from the direction opposite to that of the insulating substrate **11**. By irradiating the laser light **50P** to the first mounting pad **25a-1**, the energy of the laser light **50E** is absorbed by the first mounting pad **25a-1** and the first mounting pad **25a-1** is sublimated and removed.

[0085] In step **S12**, although FIG. **10** shows an example of retaining part of the electrode between the red LED chip **202R** and the first mounting pad **25a-1** without irradiating the laser light to the first mounting pad **25a-1** arranged in the contact hole **262**, the laser light may be irradiated to the entire first mounting pad **25a-1** **50P** to remove the first mounting pad **25a-1** so as not to interfere with the newly mounted LED **402**. In this case, the remaining part of the electrode can also be removed by irradiating the laser light **50P** in step **S12**.

[0086] The laser light **50P** may be selected as a laser light that is absorbed by the first mounting pad **25a-1**. In the present embodiment, for example, infrared or visible light centered around the wavelength of **800** nm may be used as the laser light **50P**. The laser light **50P** may be an ultrashort pulsed laser such as a femtosecond laser. The laser light **50P** may be a laser light with a different wavelength than the laser light **50E**.

[0087] Further, in step **S13** of FIG. **8**, the insulating layer **28** that was overlapped by the first

mounting pad **25a-1** that was removed in step S12. For example, the insulating layer **28** that was removed is overlapped by the first mounting pad **25a-1** is irradiated with a laser light **50C**, and the insulating layer **28** is removed. This process exposes the second mounting pad **25a-2** that was overlapped by the first mounting pad **25a-1**, as shown in FIG. 12. As shown in FIG. 11, the insulating layer **28** that was overlapped by the first mounting pad **25a-1** is irradiated with the laser light **50P** from the direction opposite to that of the insulating substrate **11**. By irradiating the laser light **50C** to the insulating layer **28**, the energy of the laser light **50C** is absorbed by the insulating layer **28** and the insulating layer **28** is removed.

[0088] The laser light **50C** may be selected as the laser light that is absorbed by the insulating layer **28**. In the present embodiment, for example, far-infrared light can be used as the laser light **50C**. A gaseous laser such as a carbon dioxide (CO.sub.2) laser can be used as the laser light **50C**. The laser light **50C** can be of a different wavelength than the laser light **50E** and the laser light **50P**.

[0089] Finally, in step S14 of FIG. 8, a new LED chip **402** is mounted on the second mounting pad **25a-2** exposed by step S13. It is sufficient to use a well-known method of mounting LED chips to mount the new LED chip **402**. For example, to mount the new LED chip **402**, a conductive bonding layer **406** may be provided on the second mounting pad **25a-2** in advance, and the first connecting electrode **303a** of the new LED chip **402** may be bonded to the conductive bonding layer **406**. As shown in FIG. 13, the new LED chip **402** can be electrically connected to the second mounting pad **25a-2** through the conductive bonding layer **406**, with the first connecting electrode **303a**, terminal electrode **403a**, and terminal electrode **403b** provided in advance.

[0090] A conductive bonding material may be used as the conductive bonding layer **406**.

Conductive bonding materials such as solder, silver paste, anisotropic conductive film (ACF), etc., such as tin (Sn) or alloys containing tin may be used.

[0091] As described above, the new LED chip **402** can be mounted in the same pixel **110** in place of the red LED chip **202R** that was determined to be defective on the circuit substrate **100**. When a plurality of LED chips **202** are provided in the pixel **101** and one red LED chip **202R** among them is determined to be defective and an LED chip **402** is mounted in place of that red LED chip **202R**, as shown in FIG. 14, the one red LED chip **202R**, the first mounting pad **25a-1** and the insulating layer **28** are removed in the repair region on the second mounting pad **25a-2**, and the LED chip **402** is mounted on the second mounting pad **25a-2** in place of the one red LED chip **402**. The region where the insulating layer **28** has been removed is shown as region **28a**.

[0092] As described above, in a repair method for a display device according to the present embodiment, by removing the LED chip **202** determined to be defective, and the first mounting pad **25a-1** overlapping the LED chip **202** and the insulating layer **28**, the second mounting pad **25a-2** overlapping the first mounting pad **25a-1** may be exposed, and a new LED chip **402** may be mounted on the second mounting pad **25a-2**. In a plan view, the second mounting pad **25a-2** can be used as a spare mounting pad without providing an additional region for arranging the second mounting pad **25a-2** in the pixel **110**, by mounting a new LED chip **402** on the second mounting pad **25a-2** that overlaps the first mounting pad **25a-1**.

[0093] Each of the embodiments described above as an embodiment of the present invention can be appropriately combined and implemented as long as they do not contradict each other. The addition, deletion, or design change of components as appropriate, or the addition, deletion, or condition change of processes by those skilled in the art based on each embodiment are also included in the scope of the present invention as long as they are provided with the gist of the present invention.

[0094] Further, it is understood that, even if the advantageous effect is different from those provided by each of the above-described embodiments, the advantageous effect obvious from the description in the specification or easily predicted by the person skilled in the art is apparently derived from the present invention.

Claims

1. A display device comprising: a first LED chip arranged in a pixel; a first mounting pad arranged in the pixel and connected to the first LED chip; a second mounting pad arranged in the pixel, overlapping the first mounting pad in a plan view and electrically connected to the second mounting pad, and an insulating layer between the first mounting pad and the second mounting pad.
 2. The display device according to claim 1, wherein a thickness of the insulating layer is greater than or equal to 1 μm and less than or equal to 2 μm .
 3. The display device according to claim 1, wherein the first mounting pad comprises a first n-electrode side pad and a first p-electrode side pad corresponding to the first LED chip, the second mounting pad comprises a second n-electrode side pad overlapping the first n-electrode side pad and a second p-electrode side pad overlapping the first p-electrode side pad in a plan view, the first n-electrode side pad is electrically connected to the second n-electrode side pad, and the first p-electrode side pad is electrically connected to the second p-electrode side pad.
 4. The display device according to claim 1, wherein the pixel comprises the first LED chip, the first mounting pad, the insulating layer, and a plurality of areas overlapped by the second mounting pad in a plan view, a repair region is included where the first LED chip, the first mounting pad and the insulating layer were removed on the second mounting pad in at least one area in the plurality of areas, and in the repair region, a second LED chip is electrically connected to the second mounting pad in place of the first LED chip.
 5. A repair method for a display device, the method comprising: irradiating a first laser light to a first LED chip determined to be defective in a pixel and removing the first LED chip; irradiating a second laser light to a first mounting pad connected to the first LED chip in the pixel and removing the first mounting pad; irradiating a third laser light to an insulating layer overlapping the first mounting pad, removing the insulating layer, and exposing a second mounting pad overlapping the first mounting pad in a plan view, and mounting a second LED chip on the second mounting pad.
 6. The repair method according to claim 5, further comprising: wavelengths of the first laser light, the second laser light, and the third laser light are different.
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